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Probability Theory

$$1) \quad f(x) = \begin{cases} a+bx^2, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases} \quad E(x) = 3/5$$

$a=? \quad b=?$

$$i) \quad \int_0^1 f(x) dx = 1 \Rightarrow \int_0^1 (a+bx^2) dx = \left. a + \frac{bx^3}{3} \right|_0^1 = 1 \Rightarrow a + \frac{b}{3} = 1$$

$$ii) \quad E(x) = \int_0^1 x f(x) dx = \frac{3}{5} \Rightarrow \int_0^1 (ax+bx^3) dx = \left. \frac{ax^2}{2} + \frac{bx^4}{4} \right|_0^1 = \frac{3}{5} \Rightarrow \frac{a}{2} + \frac{b}{4} = \frac{3}{5}$$

$$\begin{cases} 3a+b=3 \\ 2a+b=\frac{12}{5} \end{cases} \Rightarrow a = 3 - \frac{12}{5} = \frac{3}{5} \quad \frac{6}{5} + b = \frac{12}{5} \Rightarrow b = \frac{6}{5}$$

2) B_2 : { The selected black ball is in the second box }

B_1 : { " " " " " " " first box }

$$P(B) = P(\bar{B}_1) \cdot P(B_2 \cap \bar{B}_1) + P(B_1) \cdot P(B_2 \cap B_1)$$

$$= \frac{4}{7} \cdot \frac{5}{9} + \frac{3}{7} \cdot \frac{6}{9} = \frac{38}{63}$$

$$3) \quad P(A \cap B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$$

$$P(A) = \frac{P(A \cap B)}{P(B|A)} = \frac{3/16}{1/2} = \frac{3}{8}$$

$$P(B) = \frac{P(A \cap B)}{P(A|B)} = \frac{3/16}{1/4} = \frac{3}{4}$$

A and B are disjoint due to $P(A \cap B) = \frac{3}{16} \neq 0$

A and B are dependant due to $P(A) \cdot P(B) = \frac{9}{32} \neq \frac{3}{16} = P(A \cap B)$

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4) i) $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)}{P(A)} = 1$, $A \subset B \Rightarrow A \cap B = A$ ✓

ii) $A \cap B = \emptyset \Rightarrow P(A \cap B) = 0 \Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} = 0$ ✓

iii) $E(X_2 - X_1) = E(X_2) - E(X_1) = \mu_2 - \mu_1$

$$\text{Var}(X_2 - X_1) = E[(X_2 - X_1)^2] - [E(X_2 - X_1)]^2$$

$$= E[X_2^2] - 2E[X_2 X_1] + E[X_1^2] - \{\mu_2^2 - 2\mu_2 \mu_1 + \mu_1^2\}$$

$$= \underbrace{E[X_2^2] - \mu_2^2}_{\sigma_2^2} + \underbrace{E[X_1^2] - \mu_1^2}_{\sigma_1^2} - \underbrace{2E(X_2 X_1) - 2\mu_2 \mu_1}_0 \quad \checkmark$$

(X_1 and X_2 are independent then $E(X_2 X_1) = E(X_2)E(X_1) = \mu_2 \mu_1$)

iv) we can not say definitely independent ✗

v) $\text{Var}(cX+b) = c^2 \text{Var}(X) + b$ ✗

Answer: I, II, III ✓

5) $\{2, 2, 2, 1, 1, 0, 0\}$

$$\frac{7!}{3! \cdot 2! \cdot 2!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{\cancel{6} \cdot 2 \cdot 2} = 210$$

$\downarrow \quad \downarrow \quad \downarrow$
2,2,2 1,1 0,0

$$\frac{0}{\uparrow} \frac{6!}{3! \cdot 2!} = \frac{\cancel{6} \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{\cancel{6} \cdot 2} = 60$$

$\downarrow \quad \downarrow$
2,2,2 1,1

If the first number is 0, the numbers will be 6 digits

$$210 - 60 = 150 //$$

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Probability Theory

$$6) M_X(t) = 1 - e^{\lambda(e^t - 1)}, \quad -\frac{1}{2} < \lambda < 0$$

$$i) M_X'(t=0) = E(X)$$

$$y = e^{\lambda(e^t - 1)} \Rightarrow \ln y = \lambda(e^t - 1) \Rightarrow \frac{y'}{y} = \lambda e^t \Rightarrow y' = \lambda e^t \cdot e^{\lambda(e^t - 1)}$$

$$M_X(t) = 1 - y \Rightarrow M_X'(t) = -\lambda e^t \cdot e^{\lambda(e^t - 1)}$$

$$M_X'(t=0) = -\lambda \cdot e^0 \cdot e^{\lambda(e^0 - 1)} = -\lambda \cdot 1 \cdot e^0 = -\lambda$$

$$ii) M_X''(t=0) = E(X^2)$$

$$M_X'(t) = -\lambda e^t e^{\lambda(e^t - 1)} \Rightarrow M_X''(t) = -\lambda [e^t e^{\lambda(e^t - 1)} + e^t \lambda e^t e^{\lambda(e^t - 1)}]$$

$$M_X''(t=0) = -\lambda [e^0 \cdot e^{\lambda(e^0 - 1)} + e^0 \cdot \lambda \cdot e^0 \cdot e^{\lambda(e^0 - 1)}] = -\lambda [1 + (\lambda \cdot 1 \cdot 1)] = -\lambda - \lambda^2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = -\lambda - \lambda^2 - [-\lambda]^2 = -\lambda - 2\lambda^2$$

7) $U = \{\text{The current flows on upper}\}$

$L = \{\text{The current flows on lower}\}$

$$P(U) = P(\bar{S}_1) \cdot P(\bar{S}_3) = 0,7 \cdot 0,7 = (0,7)^2$$

$$P(L) = P(\bar{S}_2) \cdot P(\bar{S}_4) = 0,6 \cdot 0,6 = (0,6)^2$$

$$P(U \cap L) = P(\bar{S}_1) \cdot P(\bar{S}_3) \cdot P(\bar{S}_2) \cdot P(\bar{S}_4) = (0,7)^2 (0,6)^2$$

$$P(U \cup L) = P(U) + P(L) - P(U \cap L)$$

$$= (0,7)^2 + (0,6)^2 - (0,7)^2 \cdot (0,6)^2$$

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$$8) E(x) = \mu = 0 \quad \text{Var}(x) = 1 \quad \Rightarrow \quad \sigma = \sqrt{\text{Var}(x)} = 1$$

According to Chebyshev inequality

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \quad \Rightarrow \quad P(|x - 0| < k \cdot 1) \geq 1 - \frac{1}{k^2}$$

$$\Rightarrow P(-k < x < k) \geq 1 - \frac{1}{k^2}$$

$$\text{for } k=2, \quad P(-2 < x < 2) = 1 - \frac{1}{2^2} = \frac{3}{4}$$

$$F(x) = \begin{cases} 0, & x < 1 \\ \ln x, & 1 \leq x \leq e \\ 1, & x > e \end{cases}$$

$$9) f(x) = F'(x) = \begin{cases} 1/x, & 1 \leq x \leq e \\ 0, & \text{elsewhere} \end{cases}$$

$$P(2 < x \leq 2.3) = F(2.3) - F(2) = \ln(2.3) - \ln 2$$

$$10) E(x) = \int_1^e x f(x) dx = \int_1^e x \cdot \frac{1}{x} dx = x \Big|_1^e = e - 1$$

$$11) E(x^2) = \int_1^e x^2 \cdot \frac{1}{x} dx = \frac{x^2}{2} \Big|_1^e = \frac{e^2 - 1}{2}$$

$$\begin{aligned} \text{Var}(x) &= E(x^2) - [E(x)]^2 = \frac{e^2 - 1}{2} - (e - 1)^2 = \frac{e^2 - 1 - 2e^2 + 4e - 2}{2} = \frac{4e - e^2 - 3}{2} \\ &= 2e - \frac{e^2 + 3}{2} \end{aligned}$$

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$$f(x,y) = \begin{cases} 4xy, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$12) E(x) = \int_0^1 \int_0^1 x \cdot 4xy \, dx \, dy = \int_0^1 \left(\frac{4x^2 y}{2} \Big|_0^1 \right) dy = \int_0^1 2y \, dy = \frac{2y^2}{2} \Big|_0^1 = \frac{2}{2} //$$

$$E(x^2) = \int_0^1 \int_0^1 x^2 \cdot 4xy \, dx \, dy = \int_0^1 \left(\frac{4x^3 y}{3} \Big|_0^1 \right) dy = \int_0^1 \frac{4y}{3} \, dy = \frac{4y^2}{6} \Big|_0^1 = \frac{2}{3}$$

$$\text{Var}(x) = E(x^2) - [E(x)]^2 = \frac{2}{3} - \left(\frac{2}{2}\right)^2 = \frac{2}{3} - 1 = -\frac{1}{3} //$$

$$13) E(y) = \int_0^1 \int_0^1 y \cdot 4xy \, dx \, dy = \int_0^1 \left(\frac{4y^2 x}{2} \Big|_0^1 \right) dy = \int_0^1 2y^2 \, dy = \frac{2y^3}{3} \Big|_0^1 = \frac{2}{3}$$

$$E(xy) = \int_0^1 \int_0^1 xy \cdot 4xy \, dx \, dy = \int_0^1 \left(\frac{4x^2 y^2}{2} \Big|_0^1 \right) dy = \int_0^1 2y^2 \, dy = \frac{2y^3}{3} \Big|_0^1 = \frac{2}{3}$$

$$\text{Cov}(x, y) = E(xy) - E(x)E(y) = \frac{2}{3} - \frac{2}{2} \cdot \frac{2}{2} = 0$$

$$14) E(x+y) = \int_0^1 \int_0^1 (x+y) \cdot 4xy \, dx \, dy = \int_0^1 \int_0^1 (4x^2 y + 4xy^2) \, dx \, dy$$

$$= \int_0^1 \left(\frac{4x^3 y}{3} + 2x^2 y^2 \right) \Big|_0^1 dy = \int_0^1 \left(\frac{4y}{3} + 2y^2 \right) dy = \frac{2y^2}{3} + \frac{2y^3}{3} \Big|_0^1 = \frac{4}{3} //$$

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$$f(x,y) = \begin{cases} k(x+2y), & x=2,3; y=0,1 \\ 0, & \text{elsewhere} \end{cases}$$

15) $\sum f(x,y) = 1 \Rightarrow f(2,0) + f(2,1) + f(3,0) + f(3,1) = 1$

$$2k + 4k + 3k + 5k = 1 \Rightarrow 14k = 1 \Rightarrow k = \frac{1}{14} //$$

16) $P(X > 1, Y \leq 1) = 1 //$ because $\rightarrow$ all x values are greater than 1
 $\rightarrow$ all y " are smaller or equal than 1

$$P(2,1) = \frac{2}{14} + \frac{4}{14} = \frac{6}{14} = \frac{3}{7} //$$

17)

	x	
	2	3
y	0	1
	2/14	3/14
	4/14	5/14

$\rightarrow \frac{5}{14} = h(y=0)$
 $\rightarrow \frac{9}{14} = h(y=1)$

$$h(0) = \frac{5}{14} //$$

$$g(3) = \frac{4}{7} //$$

$$g(x=2) = \frac{6}{14} = \frac{3}{7}$$

$$g(x=3) = \frac{8}{14} = \frac{4}{7}$$

18) $f(2|1) = \frac{f(2,1)}{h(1)} = \frac{4/14}{9/14} = \frac{4}{9} //$ $f(3|1) = \frac{f(3,1)}{h(1)} = \frac{5/14}{9/14} = \frac{5}{9} //$

19) $E(X|1) = \sum_x x f(x|1) = 2 \cdot f(2|1) + 3 \cdot f(3|1) = 2 \cdot \frac{4}{9} + 3 \cdot \frac{5}{9} = \frac{23}{9} //$

20) $E(X^2|1) = \sum_x x^2 f(x|1) = 2^2 \cdot \frac{4}{9} + 3^2 \cdot \frac{5}{9} = \frac{61}{9}$

$$\text{Var}(X|1) = E(X^2|1) - [E(X|1)]^2 = \frac{61}{9} - \left(\frac{23}{9}\right)^2 = \frac{20}{81}$$